

15. Evie and Xavier are playing a game rolling a fair standard six-sided die. Evie wins if she rolls a 6 before Xavier rolls an odd number. If Evie rolls first, determine the probability Evie wins the game. Express your answer as a common fraction reduced to lowest terms.
16. Henrik measured the angle of inclination from the ground to the top of a vertical pole along a line on level ground and found the angle to be 50° . He walked 10 feet further from the pole along the same line and measured the angle of inclination from ground level to the top of the pole to be 40° . If he walked along the same line to a point 10 feet from the base of the pole, the angle of inclination from ground level to the top of the pole is k° . Determine k . Express your answer as a decimal rounded to the nearest hundredth of a degree.
17. Determine the point on the graph of the function $y = \sqrt{2x-5}$ that is closest to the point $(5,0)$. Express your answer as an ordered pair with exact coordinate entries.
18. When written out as an integer, $2015!$ has k trailing zeros (2015 factorial ends in a string of k zeros.) Determine the value of k .
19. Let $\sec \theta = \frac{20}{15}$. Determine the least possible exact value for $\sin \theta$.

20. In the diagram shown, $\overline{CD}$ bisects $\angle ACB$ in $\triangle ABC$.
 $\cos \theta = \frac{1}{4}$ and $\frac{1}{a} + \frac{1}{b} = \frac{1}{20}$. Determine the exact length CD .

